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LiSA-MobileNetV2: an extremely lightweight deep learning model with Swish activation and attention mechanism for accurate rice disease classification

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In the context of intelligent agriculture in China, rapid and accurate identification of crop diseases is essential for ensuring food security and improving crop yield. Although lightweight convolutional neural networks (CNNs) are widely adopted for plant disease recognition due to their computational efficiency, they often suffer from limited feature representation and classification accuracy. To address these challenges, we propose LiSA-MobileNetV2, an improved MobileNetV2-based model designed for rice disease classification. First, we restructure the inverted residual module to simplify the network architecture, achieving a test accuracy of 92.32%, representing a 2.41% improvement over the original MobileNetV2 (89.91%). This indicates that a more lightweight network can enhance feature representation in specific disease recognition. Second, integrating the Swish activation function further improves accuracy to 94.04% by enhancing the model's non-linear feature learning. Finally, the addition of a squeeze-and-excitation attention mechanism raises accuracy to 95.68%, representing a 5.77% improvement over the original model. Importantly, the parameter size and FLOPs are reduced by 74.69% and 48.18%, respectively, maintaining strong computational efficiency. These results demonstrate that combining structural simplification, advanced activation, and efficient attention mechanisms significantly improves CNN performance. LiSA-MobileNetV2 provides a high-accuracy, resource-efficient solution for real-time rice disease detection in smart farming systems.

KEYWORDS

disease recognition, rice, lightweight CNN, attention mechanism, activation function